

# *cs-mail* Deployment and Migration Profile

## Initial C-SQD Operation, Funding, SMTP, and Clearing

C-SQD

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### Abstract

This document describes nonnormative choices for deploying the *cs-mail* protocol. It separates the protocol’s invariant meaning from operator policy: identity enrollment, account funding, risk controls, SMTP migration, provider clearing, privacy operations, and staged launch. The profile gives one plausible path for an initial centralized C-SQD service and later federation. It neither changes the settlement rules in the *cs-mail Protocol Specification* nor requires another provider to adopt the same vendors, verification evidence, storage topology, or banking rails.

## 1 Status and Document Boundary

The *cs-mail Protocol Specification* is normative. It defines permission, bonds, repeated-attempt controls, canonical ordering, settlement, privacy, and invariants. The *Rust Reference Architecture* explains one implementation shape. This profile addresses choices that a real operator must make but that do not belong in the portable protocol.

In particular, this profile does not define a universal legal identity, mandate bank-based enrollment, select a payment processor, create an additional protocol fee, or weaken native end-to-end content encryption. Policies in this document must remain versioned, disclosed, auditable, and subordinate to protocol semantics.

## 2 Initial Deployment Shape

An initial deployment may place several logical roles under C-SQD while keeping their authority and data access separate. C-SQD may operate decision ingress, the canonical journal, relationship state, the internal ledger, native delivery, and internal clearing. External regulated providers may fund and redeem user balances.

Central operation does not justify a universal database credential or identifier. Before meaningful user data accumulates, the deployment should establish separate access domains for:

- principal evidence and account recovery;
- public identities and directed relationships;
- encrypted message content;
- balances, bonds, reserves, and settlement;
- delivery and abuse telemetry; and

- separately required compliance or dispute records.

Each domain should have purpose-specific credentials, encryption keys, retention classes, and access audit. Translation among scoped references is an explicit operation. A central operator should not treat organizational ownership as a substitute for technical compartmentalization.

### 3 Identity Enrollment and Principal Policy

The protocol requires a continuing private principal for control, recovery, and repeated-attempt continuity. It does not require that every provider define that principal as a legally verified natural person. C-SQD should publish which kinds of principal it supports, what evidence each tier requires, and which functions depend on verification.

One deployment might accept several enrollment profiles:

| Profile         | Possible evidence                                                                                                                                   | Policy considerations                                                                                                           |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Individual      | Account authentication, recovery factors, and, where justified, legal name, date of birth, or a verified funding account.                           | Avoid collecting legal identity merely because it is conventional; document necessity and retention for each field.             |
| Organization    | Organization name, jurisdiction or registry reference, institutional funding account, and evidence that an enrolling representative may act for it. | Support delegated administration, representative turnover, and recovery without making registry data part of ordinary messages. |
| Managed group   | Sponsoring organization, administrator authority, and member or role controls.                                                                      | Define who owns protocol identities, receives funds, and may recover or revoke operational keys.                                |
| Limited account | Provider-local account and strong recovery evidence without conventional bank or legal-identity verification.                                       | Apply proportionate balance, withdrawal, or sending limits without exposing a public identity tier.                             |

Exact evidence is a legal, risk, accessibility, and inclusion decision. People who cannot or should not use conventional bank-based verification need a documented path where feasible. Identity evidence and the provider-local principal reference must never appear in ordinary message envelopes, relationship records exposed to other users, or globally comparable federation assertions.

Provider migration should use recipient-scoped continuity proofs for repeated- attempt enforcement. It should not export a stable universal principal identifier.

### 4 Funding, Withdrawal, and Financial Risk

Users may fund ledger balances periodically through cards, bank transfers, or other supported rails. Protocol settlement then occurs internally. A valid bond must remain fully backed even if an external provider later delays, reverses, or disputes a funding transaction.

The operator therefore needs explicit policy for:

- funding finality and any waiting period before deposited value may back a bond;
- withdrawal availability for returned collateral and received settlement;
- chargebacks, fraud losses, negative balances, account suspension, and separately authorized compensating ledger entries;

- limits by verification tier, settlement unit, jurisdiction, or risk;
- sanctions, tax, money-transmission, consumer-protection, and abandoned- property obligations where applicable; and
- dispute windows and retention classes for evidence that must survive longer than ordinary delivery data.

External reversals do not rewrite a terminal protocol event. If policy authorizes a later adjustment, the ledger records a new compensating batch with its own cause, authority, and audit trail.

The sender-facing interface should distinguish available, reserved, pending- release, and withdrawable value. Before authorization it should show the full  $C + S + L_k$  requirement, the short admission window, the decision-window duration that begins only after admission, the earliest final expiry time, and the persistence-reserve release condition. A still-unadmitted reservation should offer cancellation and explain that cancellation or admission timeout returns all reserved value.

## 5 SMTP Migration

SMTP is an edge compatibility mechanism, not the native trust or privacy model. The gateway should make the route and its privacy consequences visible.

### 5.1 Inbound Legacy Sender

When a conventional sender contacts a C-SQD recipient, the gateway can follow this sequence:

1. Resolve whether the legacy sender maps to an accepted protocol identity.
2. If accepted-sender authority exists, apply ordinary accepted-sender and safety policy.
3. Otherwise, return a standards-compatible challenge or a web link that explains the recipient's current contact terms.
4. Authenticate or enroll the sender, present the complete reservation, and fund the sender account if necessary.
5. Obtain fresh signed terms and reserve the bond and any persistence reserve with a short admission deadline.
6. Durably record the encrypted content and retryable delivery intent, then admit the attempt through the same deterministic protocol command used by a native client. If this does not happen before the admission deadline, return  $C + S + L_k$  in full.
7. Return an authenticated admission result without implying inbox rank, recipient attention, or a guaranteed reply.

The web flow must bind the funded authorization to the exact sender, recipient, message, quote, amount, admission deadline, and later decision deadline. A reusable payment link or bearer token must not let another party substitute content or redirect a reservation.

A legacy gateway may encounter plaintext before creating native ciphertext. The message must therefore be marked as outside the native endpoint-to-endpoint confidentiality guarantee. Transport TLS and provider encryption at rest are not equivalent to native end-to-end encryption.

## 5.2 Outbound Legacy Recipient

When a native user sends to a conventional recipient, the client should disclose that the destination cannot enforce native permission, settlement, graph privacy, or content confidentiality. The envelope and local history should record the downgrade. The system must not advertise a successful SMTP delivery as recipient acceptance of a *cs-mail* relationship.

## 5.3 Gateway Abuse Controls

A payment challenge is itself a communication surface. Gateways should rate-limit challenges, prevent address enumeration, avoid leaking recipient pricing to unauthorized bulk probes, expire unfunded flows, and provide a way to report impersonation or malicious content. Gateway policy may reject malware, unlawful content, or excessive volume independently of financial validity.

# 6 Provider Clearing

Federation creates obligations between providers when recipient providers earn  $C$  or recipients receive  $S$ . The persistence reserve  $L_k$  remains sender property and never enters provider revenue or inter-provider clearing.

Providers may preserve each authenticated protocol event while periodically netting external payment. Suppose Provider A owes Provider B 10,000 during a clearing period and Provider B owes Provider A 8,000. The external rail may settle a 2,000 net obligation from A to B. The underlying events remain individually attributable through scoped settlement receipts so participants can verify their balances and authorized disputes can be resolved.

A clearing agreement should define:

- supported settlement units and no implicit foreign-exchange rule;
- clearing intervals, collateral or credit limits, and default handling;
- canonical cutoffs and treatment of late or disputed events;
- reconciliation proofs and scoped audit references;
- who bears external-rail fees without changing quoted protocol settlement; and
- suspension and orderly wind-down without stranding participant-owned balances or pending protocol decisions.

Netting changes how providers discharge obligations. It does not change who owns value after any acceptance, rejection, expiry, or unadmitted-cancellation transition.

# 7 Operational Policy and User Experience

The recipient and recipient provider have economic incentives that require visible counter-controls. C-SQD should publish bounds or review procedures for  $C$  and provider minimum collateral, prohibit interfaces that hide timely acceptance, issue signed decision receipts, detect recipient-provider collusion, and provide dispute and suspension procedures.

The product should teach relationship consequences at the decision point. An accept, reject, or block control beside a message should state that the action applies to the directed sender identity,

settles every decision-open admitted bond for that relationship, and fully cancels its unadmitted reservations. Acceptance releases persistence reserves only for attempts from that public identity; it does not reset principal-level backoff or release reserves belonging to another identity of the same principal. Rejection declines the active solicitation while preserving a later eligible retry after backoff. Blocking is the separate control for prohibiting future ordinary contact until the recipient removes the block. A late acceptance should distinguish prospective permission from older bonds whose decision windows are already closed.

The sender-facing flow should distinguish reservation from admission. Closing the client, cancelling, or failing to make encrypted content and its delivery intent durable before the admission deadline returns  $C + S + L_k$  in full, advances no backoff, and produces no recipient solicitation. The recipient decision window starts only at canonical admission.

The first admitted attempt for an unaccepted directed relationship should open one grouped relationship solicitation. That group is the one logical establishment prompt; recipient settings may present it as a push, badge, banner, inbox row, or quiet equivalent. Later admitted messages join that active group. They may update its contents or ordinary unread count quietly, but must not repeatedly ask the recipient to establish the same relationship. This behavior is solicitation coalescing, not message idempotency: every attempt retains its own bond, deadline, content, and repeated-attempt consequence.

Accept, reject, and block close the active solicitation and apply to the relationship. Reject permits a later eligible attempt after the applicable backoff. Block refuses future terms, reservations, any optional capability use, and admission until the recipient unblocks; unblocking does not accept the sender or erase backoff. Ignoring the solicitation means taking no action, so bonds still expire individually. Dismissing or hiding the prompt is a presentation preference and must not be represented as acceptance, rejection, blocking, or financial settlement. Once the final associated bond is no longer decision-open, the episode may lapse. A later eligible attempt may open one new episode after the applicable backoff unless the relationship is blocked.

Operational dashboards should report service health without reconstructing a global relationship graph. Metrics should use short windows, aggregation, scoped trace references, and documented retention. Support access to identity, relationship, content, or ledger data should be separately authorized rather than granted through one universal administrator role.

## 8 Staged Launch

Express lanes are a launch extension outside settlement state. Their recipient- signed grants authorize only rate- and time-bounded bond-free communication, open no relationship solicitation, advance no repeated-attempt state, and remain subordinate to recipient blocking.

A cautious deployment can proceed in five gates:

1. **Closed deterministic kernel.** Execute every transition and invariant in memory using fixed policy and explicit canonical time.
2. **Internal-value pilot.** Add the append-only ledger, transactional outbox, crash recovery, and synthetic or nonwithdrawable test value.
3. **Native-client pilot.** Add endpoint encryption, key transparency, recovery, privacy-domain credentials, retention, and verified deletion.
4. **Limited external funding and SMTP.** Introduce real rails and gateways under conservative limits, visible privacy downgrades, and staffed dispute operations.

5. **Federation.** Add independent providers only after golden transition vectors, authenticated receipts, clearing reconciliation, migration proofs, and adversarial failure tests are stable.

Each gate should have measurable exit criteria. Volume growth is not evidence that value conservation, privacy deletion, decision timeliness, or dispute operations work correctly.

## 9 Conclusion

The first C-SQD deployment can be centralized without treating deployment policy as protocol law. Identity evidence, funding rails, SMTP behavior, risk controls, and clearing agreements should remain explicit profiles around the deterministic kernel. Preserving that boundary allows the product to evolve, jurisdictions to differ, and later providers to interoperate without changing the meaning of permission or settlement.